

URJITA BEDEKAR

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RESEARCH INTERESTS

Benchmarking and evaluation of data systems for AI – measuring whether a data or system intervention genuinely helps, and building rigorous, contamination-resistant benchmarks for ML and compound-AI pipelines at production scale. Interested in workload synthesis for evaluation, data-centric methods that improve model reliability, and the validation problems created by cheap synthetic generation.

EDUCATION

M.S., Computer Science, Rochester Institute of Technology, Rochester, NY – 2021 – 2024

Coursework: Machine Learning · Foundations of AI · Information Retrieval · Big Data · Database System Implementation · Non-Relational Data Management

RESEARCH EXPERIENCE

Graduate Capstone Research, Rochester Institute of Technology – 2024

Advisor: Dr. Weijie Zhao

- Designed a controlled evaluation to determine *which* data interventions genuinely improve a model under severe class imbalance – raising accuracy from **88%→95%** (XGBoost) and **89%→94%** (RandomForest) across a 50,425-record corpus of four unevenly distributed categories.
- Established that random insertion, swap, and deletion *degraded* performance by disrupting the syntactic structure carrying class signal – a negative result that motivated selecting only the augmentations preserving category-discriminative features.
- Built the evaluation pipeline enabling like-for-like comparison across data conditions: SpaCy tokenization and lemmatization, TF-IDF vectorization, SMOTE, controlled augmentation.

Machine Learning Research Assistant, Rochester Institute of Technology – 2022

- Investigated EEG-derived neural biomarkers across **70+ participants**, developing an end-to-end signal-processing methodology (FIR filtering, ICA artifact removal, denoising, epoching, normalization) to render raw neural recordings into rigorously validated ML inputs.
- Engineered time-frequency representations (power spectral density, ERPs, alpha/beta rhythms, spectral entropy, coherence) and evaluated SVM, random-forest, and neural-network classifiers, validating results with cross-validation, ROC analysis, and ANOVA rather than accuracy alone.

AI for Earth Grantee, Microsoft – 2018 – 2019

- Co-authored *Rehabilitation and Protection of Coastal Areas*, a competitively funded research proposal selected as **1 of 236 worldwide** for a \$15,000 Azure grant.
- Developed **convolutional neural network** models for deforestation detection from satellite imagery, achieving automated land-cover classification across **50+ global regions** (OpenCV, TensorFlow, scikit-learn).
- Eliminated manual data collection by automating the end-to-end satellite image-capture and ingestion pipeline with **UiPath, Docker, and Kubernetes**, keeping the models continuously retrained on the latest imagery.

Data Scientist (Internship), CSIR Central Drug Research Institute – 2019

- Investigated machine-learning methods for early Parkinson's detection, leading a five-person team to construct the dataset and benchmark classifiers (logistic regression, KNN, random forest) to **93% accuracy**.
- Automated dataset construction across **100+ external sources** (Python, Selenium, BeautifulSoup), replacing manual collection entirely.

PROFESSIONAL EXPERIENCE

Google · Technical Program Manager, Search Infrastructure · 2025

- Built an ML forecasting system that predicts compute/storage demand from historical telemetry and flags resource over-subscription before launch, freeing \$1M+ in CapEx per major release across 200+ infrastructure roll-outs.
- Defined telemetry data models and evaluation criteria (SLOs/SLIs, launch gates) that surfaced reliability anomalies 30% faster and cut manual operational work 80%.

Tesla · Engineering Co-op · 2022, 2023

- Cut production data latency from 10 minutes to under 10 seconds via automated data validation, logging, and monitoring across pipelines serving 250K+ vehicles.
- Built ETL pipelines and REST services (Python, MySQL, Django) integrating 5+ operational systems into one normalized data layer for downstream analytics and ML workflows.

Accenture · Software Engineer · 2021

- Automated CI/CD pipelines (Azure DevOps, Git) catching 20+ major pre-deployment errors; built C#/ASP.NET/SQL Server features and test-automation frameworks, cutting manual processing 40%.

PUBLICATIONS

- A Novel Approach to Recommend Skincare Products Using Text Analysis of Product Reviews**. ICTCS 2020 (Springer). NLP pipelines converting unstructured review text into structured signals powering a recommendation model.
- Harvest Treasure: Prediction of Best Crop Yield**. ICICN 2020 (Springer). Feature engineering over multi-dimensional environmental datasets, benchmarking supervised models in an end-to-end forecasting pipeline.

HONORS

- Microsoft AI for Earth Award**, 2019 – co-wrote the successful proposal; \$15,000 Azure compute grant, 1 of 236 projects selected worldwide.
- Highest merit-based scholarship**, Rochester Institute of Technology – 40% of M.S. tuition.

TECHNICAL SKILLS

Python, C++, C#, Java, MATLAB, R, SQL · PyTorch, TensorFlow, scikit-learn, NumPy, pandas · Spark, Hadoop, PostgreSQL, MySQL · Django, Flask, REST APIs, React.js · Docker, Kubernetes, UiPath, Linux, Git, CI/CD · Data augmentation, benchmarking, model evaluation, LLM evaluation, cross-validation, ROC analysis · ETL pipelines, data validation, time-series modelling, SLO/SLI definition